

Optimization of a Plant Microbial Fuel Cell via Inoculation of Plant Roots with Arbuscular Mycorrhizal Fungi

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Abstract – The use of fossil fuels is becoming more harmful, with increasingly pronounced climate and health effects. Hence, alternative energy sources, such as plant microbial fuel cells (PMFCs) are being developed. They produce bioelectricity when plant roots secrete root exudates (containing sugars, organic acids, and amino acids), which are broken down by electrochemically active bacteria (EAB) using a metabolic pathway, yielding electrons that move from the EAB to the anode, then to the cathode. This study assessed the effects of Arbuscular Mycorrhizal Fungi (AMF) inoculation on electricity generation in a PMFC—quantified using voltages, current densities, power densities, and polarization/power density curves. Implemented in these PMFCs were *Epipremnum aureum*, a houseplant with a high growth rate and biomass; *Bacillus subtilis*, a species of EAB successfully used in bioelectrical systems; and an affordable bentonite clay membrane, with excellent semipermeable qualities. AMF was incorporated to increase biomass and root lengths of plants, providing more substrate for microbial activity. Plant health was observed by measuring chlorophyll content and plant height. The findings show that AMF did not increase electricity production; the group with *Bacillus subtilis* only produced significantly higher voltages than the others (36.85 mV—19.8% higher on average), along with an average power density of 687.35 mW/m²; three times higher than that of previous literature [23], while the voltage was 468.15 mV lower. Thus, AMF inhibited the electricity generation of *Bacillus*, which suggests competition between *Bacillus* and AMF.

Keywords: Plant microbial fuel cell, Arbuscular mycorrhizal fungi, *Epipremnum aureum*, *Bacillus subtilis*, bioelectricity

I. INTRODUCTION

As the world develops, the environmental impacts of using fossil fuels (coal, oil, natural gas) for producing electricity become increasingly detrimental [2]. They are behind more than three-quarters of greenhouse gas emissions [20], making them one of the driving forces behind climate change: an increasingly prevalent problem with the rise of air and ocean temperatures, which leads to more droughts, floods, and heavy storms. These effects on the climate impact human health by elevating risks of heat-related death, along with increasing exposure to harmful chemicals (due to contamination of soil and water) and food/waterborne infections, according to the CDC. The burning of fossil fuels also results in air pollution, which accompanies the release of microscopic particles that lead to cardiovascular diseases and chronic asthma [13]; this accounts for 5 million premature deaths every year [14]. Even with these drawbacks, fossil fuels remain the primary energy source worldwide because they are the most efficient and well-

established [3]. Therefore, improving the efficiency and availability of alternative energy sources such as solar, hydroelectric, and wind power are necessary to mitigate the effects of fossil fuels [17] [18].

The use of clean, renewable energy has been increasing. According to the US Energy Information Administration, it is mostly used in the electric power sector and accounted for 12% of US energy consumption in 2020—up 1% from 2018. However, many power sources have drawbacks. For example, geothermal energy, which uses heat from the Earth's interior and is accessed through drilling, is unevenly distributed throughout the world [17]. Also, many people must be displaced from their homes in order to implement hydropower, which involves building dams with turbines to harness the energy of water flowing from high to low elevations [17]. In addition, the technological innovation required to implement renewable energy sources such as solar panels and hydropower can be costly—more so than using fossil fuels [17]. For example, many people must be displaced from their homes in order to implement hydropower [17]. Also, the expenses surrounding solar energy result from the use of pure silicon, which takes a great deal of energy to manufacture; however, there are incentives provided by the government for lower socioeconomic classes to obtain solar panels, which account for 80% of adopters in low to moderate income households (still making up significantly less of solar panel owners than those with high incomes) [16]. Moreover, solar and wind power are subject to seasonal variability, and the large batteries used to alleviate this problem would be costly to construct [18]. Even so, renewable energy is more accessible since it is available throughout the world—an important aspect to minimize climate change [17].

Plant microbial fuel cells (PMFCs) produce clean, renewable, and sustainable bioelectricity which—as their power generation improves—may help reduce the widespread use of fossil fuels [12] [11]. They can be used with houseplants, green roofs, and rooftop gardens; furthermore, on a larger scale, they can be incorporated into agriculture without negatively affecting food production and used in wetlands and wastelands to generate energy [11]. For example, they have successfully been employed in wetlands and paddy fields to produce bioelectricity while reducing methane emissions [11]. Furthermore, by implementing lower-cost composite ceramic/bentonite clay membranes, they can be more affordable than systems using expensive Nafion membranes [21]. However, the highest voltages of PMFCs are currently not commercially viable, and fossil fuels are significantly more efficient.

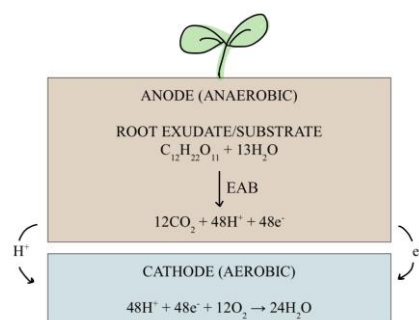


Figure 1. Reactions in a PMFC

Double-chambered PMFCs involve an anode and cathode chamber, separated by a proton exchange membrane (PEM), which allows for the diffusion of hydrogen ions to the anode side once a voltage is reached (when enough electrons move from the anode to cathode chamber) [11]. The anode chamber contains the plant, soil, and anode, while the cathode is exposed to the atmosphere. The electrochemically active bacteria (EAB), usually added into the soil intentionally, oxidize rhizodeposits (products deposited by the plant roots, including root exudates inhibiting the growth of harmful microbes and pathogens) containing sugars and organic matter, yielding protons, electrons, and CO_2 (Fig. 1). Depending on the species of EAB, this is done using mediators (external or produced by the bacteria), a direct electron transfer, or conductive nanowires that extend from the outer membrane of bacteria [15] [22]. Additionally, plant species with more rhizodeposits are generally preferred because they would provide more substrate available for EAB to oxidize at the roots [23] [11]. Having an extensive root system is also beneficial, as it allows for more microbial attachment at the roots, increasing the rate which rhizodeposits are metabolized [11]. Hence, electrons are transported to the anode, which is connected to the cathode using a wire containing the resistor/load. Using this wire, electrons flow to the cathode, while the protons go through the PEM [22]. At the cathode, the protons and electrons react with oxygen to form water [22]. It can be noted that sewage sludge, which contains a variety of microorganisms, is often used as a source of EAB in PMFCs and MFCs; however, this would be difficult to reproduce in different locations and certain bacterial species tend to contribute more than others [15].

To quantify electricity production in PMFCs, voltage, power density, current, and current density are generally recorded. Voltage (the difference in charge between two points, measured in volts) can be measured with a multimeter, and is equal to current (the rate that charge flows at, measured in amperes) multiplied by resistance (opposition to current flow, measured in ohms). Power density and current density can be found by dividing voltage and current by anode surface area [20]. There is much room for improvement for power density, current density, voltage, and current in PMFCs when compared to other energy sources like MFCs [11].

The highest voltages for PMFCs are not substantial enough to generate enough energy commercially, and their design

may limit their efficiency; therefore, further exploration/optimization is necessary [11]. When compared to other energy sources such as standard MFCs, PMFCs have not been able to perform equivalently. Moreover, the different factors each study incorporates makes it difficult to compare results between different studies. While a wide range of varying designs and materials have been tested for PMFCs, the most efficient one was found in [21] and involves the use of *Epipremnum aureum*, a composite ceramic membrane, and modified carbon fiber brush electrodes. Unlike others, it uses a novel modular triple (rather than double) chambered design, which increases servicability. This study was able to build a PMFC consistently producing voltages upwards of 800 mV for the last 25 weeks of a 32 week observation period, along with a power density and current density of 24.56 mW/m^2 and 52 mA/m^2 . Aside from the configuration, many factors contribute to its electricity generation: *E. aureum*, the plant used, was augmented with additives such as compost and dead roots to produce considerable amounts of biomass, which would provide more substrate for EAB to oxidize; carbon fiber brush electrodes were modified with acid to enhance roughness (a quality which increases microbial attachment); and a ceramic clay membrane with bentonite and flyash mixed in, which has comparable qualities to the more expensive and extensively used Nafion membranes due to its ability to allow protons to diffuse across chambers without compromising overall performance of the PMFC, was used.

Various additives to increase the bioelectricity production of PMFCs include adjusting the substrate salinity [12] and using biochar [5]. Other avenues of optimization are also being explored, such as incorporating plants with more biomass and a longer lifespan [12]. In addition, endophytes such as AMF, which have not been investigated yet, could be incorporated beneficially due to their bioremediation capacity and metabolic diversity [12]. Low bioelectricity generation is the main challenge of PMFCs, so more research concerning this must be done before their full-scale [11].

Epipremnum aureum is a viable plant for PMFC applications, as a houseplant with wide availability and low maintenance requirements, along with a high growth and development rate and significant amounts of biomass [20]. Out of the plants studied in the context of a PMFC by [20], *E. aureum* had the fastest growth rate from the second week. It has also produced a maximum voltage significantly higher than another plant (620 mV compared to 432 mV). Moreover, in a study with a triple-chambered modular PMFC using *E. aureum* and a composite ceramic membrane, voltages produced remained almost constant at around 900 mV from the 12th week onwards in a six month study [21], which helps show that PMFCs are capable of generally continuous bioelectricity generation for extended periods of time. This is important because some alternative energy sources, such as those relying on the ocean and wind, exploit a resource that fluctuates over time, which tends to result in a lack of continuous energy production [17].

Arbuscular Mycorrhizal Fungi (AMF) significantly increase root branching and microbial biomass [10], which makes it a good candidate for improving the bioelectricity generation of PMFCs [12]. It does this by increasing mobilization of photosynthates (e.g. sugars) toward plant roots, as well as increasing water and nutrient uptake. It can also form mutualistic associations with about 72% of vascular plants [4] and plays a significant role in the natural ecosystem [10]. In addition, AMF significantly increased chlorophyll A content by 36.6% and microbial biomass (in soil) by 29.0% in okra plants under drought stress [10]. An increase in microbial biomass may lead to an increase in bioelectricity output because plants with higher biomass are preferred for PMFC applications [22]. The same study also observed a change in root surface area, from $5.38 \pm 0.57 \text{ cm}^2$ in control plants without AMF to $5.94 \pm 0.09 \text{ cm}^2$ with AMF. Although this was done with a sample size of 3 plants per group, larger root surface areas may increase the amount of space for microbial attachment and rhizodeposition [11]. Such endophytes, which have not been investigated yet, have been suggested to potentially benefit electricity generation due to their bioremediation capacity and metabolic diversity [12]. This study attempts to investigate whether the added factor of AMF inoculation significantly increases power output in a PMFC. Voltage, current density, power density, and polarization and power density curves were measured and analyzed to quantify electricity generation. Plant health was assessed using chlorophyll content and plant height.

II. METHODS AND MATERIALS

2.1 Materials

Epipremnum aureum, carbon fiber cloth, and bentonite clay were purchased from Amazon. A tube of *Bacillus subtilis* was purchased from Carolina Biological, along with nutrient agar. Graphite rod anodes, ammonium chloride, potassium phosphate, sodium hydroxide, and sodium phosphate were obtained from Flinn Scientific; magnesium sulfate was obtained from Fisher Scientific; potassium phosphate was obtained from Central Scientific; and calcium chloride was obtained from ScholAR Chemistry.

2.2 Experimental design

To find the impact of AMF on the electricity generation of a PMFC, four groups were used: a control group (with the plant only), a *Bacillus* only group (with only the plant and *B. subtilis*), an AMF only group (with only the plant and AMF), and an AMF + *Bacillus* group (with the plant, *B. subtilis*, and AMF). The experiment took place over a period of 5 days, with voltage data and plant growth data being collected daily and chlorophyll content and polarization/power density curves after the experimental period. Additionally, the plants chosen to be used with each PMFC had root lengths within 5 cm of each other.

2.3 Preliminary tests

To determine a sufficient amount of bacteria to use in the anode chamber, a concentration of 10^7 CFU/mL [7] was

initially tested; since electricity generation did not change between the control group and *Bacillus* only group, it was determined to not be significant enough. A concentration of $1.2 \times 10^9 \text{ CFU/mL}$ was eventually implemented after testing increasing amounts of bacteria.

2.4 *B. subtilis* culture and media

Cultures of *B. subtilis* were streaked on nutrient agar plates at incubated at 30 degrees celsius before use. Prior to inoculation (at locations depicted in Fig. 2), anode bacteria was suspended in 10 mL sterile water [7] and adjusted to $1.2 \times 10^9 \text{ CFU/mL}$. Bacteria in the catholyte was suspended in 300 mL M9 medium ($1 \text{ g L}^{-1} \text{ NH}_4\text{Cl}$, $3 \text{ g L}^{-1} \text{ KH}_2\text{PO}_4$, $6 \text{ g L}^{-1} \text{ Na}_2\text{HPO}_4$, $5 \text{ g L}^{-1} \text{ NaCl}$, $1 \text{ mmol L}^{-1} \text{ MgSO}_4$, $0.1 \text{ mmol L}^{-1} \text{ CaCl}_2$, $0.5 \text{ mol L}^{-1} \text{ NaOH}$) [15] and adjusted to $5 \times 10^6 \text{ CFU/mL}$ [19].

To find CFU/mL, absorbances derived from McFarland standards were used and measured with a Vernier spectrophotometer: 0.530 nm for the anode chamber and 0.050 nm for the cathode chamber [9]. To make the suspensions, *B. subtilis* from agar plates was put into water or M9 medium, and concentrations were adjusted using the McFarland standards as an indicator for when the desired CFU/mL was reached.

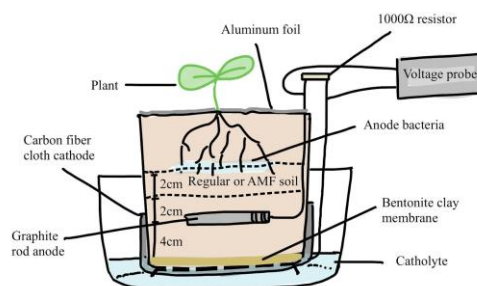


Figure 2. PMFC construction

2.5 PMFC construction

In this study, one PMFC was constructed for each experimental group. To make them (as shown in Fig. 2), a carbon fiber cloth cathode was wrapped around the bottom of the plant pot; then, the bentonite clay membrane, made by spreading bentonite clay on wire mesh, was put into the bottom, and soil, along with the anode, were layered on top. Depending on the experimental group, AMF soil mix, composed of 25g AMF/500g dry potting soil [8], and/or bacteria were added. Finally, the plant was placed in and aluminum foil covered the soil, keeping the anode chamber anaerobic (depicted in Fig. 3). This pot was put into a container containing the 300 mL of catholyte, which was replenished as necessary throughout the course of the experiment.

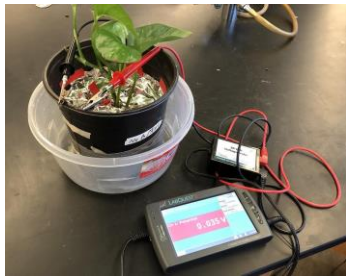


Figure 3. Experimental setup with voltage probe and LabQuest

2.6 Data collection

2.6.1 Voltage

The voltages of PMFCs were measured using a Vernier LabQuest attached to a Vernier voltage probe (Fig. 3). Voltage data was collected for 1 hour per day (represented by an average) over the period of 5 days. Average voltages, depicted in a column chart as well as a line graph showing variation of voltage over time, along with maximum voltages, were noted.

2.6.2 Power and current densities

Power and current densities were obtained by dividing the power and current (respectively), calculated using voltage and external resistance, over the anode surface area (Fig. 4) [20]. For these values, a maximum was obtained for each group, along with averages, which were graphed in a column chart.

$$\begin{array}{lll} \text{Current density:} & \text{Power density:} & V = \text{voltage} \\ I = \frac{V}{\alpha R} & P = \frac{V^2}{\alpha R} & R = \text{external resistance} \\ & & \alpha = \text{anode surface area} \end{array}$$

Figure 4. Calculations for current and power densities

2.6.3 Polarization and power density curves

Polarization and power density curves (showing voltage and power density as functions of current density) were obtained for each group in the experiment. They are typically collected by varying external resistances in a fuel cell (which would change the current density), and observing responses of voltage and power density. Polarization curves are characterized by three regions, which can provide insight to types of voltage losses the PMFC experiences. Occurring at low current densities, activation losses represent the excess electric potential needed to bypass the activation energy of anode reactions. At intermediate current densities, ohmic losses occur due to the intrinsic characteristics of conductive electrical components (such as wires), which linearly lowers voltage. Finally, at high current densities, voltages drop due to mass transport, which deals with the movement of mass (such as uncharged particles, which lead to performance losses) between locations in the PMFC. In this study, a series of loads from 100Ω-1000Ω (100, 300, 560, 750, 1000) were applied to PMFCs for 10 minutes each, and a voltage

value (along with a corresponding current and power density) was obtained at each external resistance [1] [20]. These points were then graphed, with current density on the x-axis, voltage on the y-axis, and power density on a secondary y-axis.

2.6.4 Chlorophyll content

Chlorophyll content, commonly used as an indicator of plant health [6], was also assessed. To extract the pigments, a 10 mL solution containing 0.3g of a leaf sample macerated in absolute ethyl alcohol was created then left in the dark at 4 degrees Celsius for 24 hours. After, this was centrifuged at 6500 rpm for 15 minutes. The absorbance of the resulting supernatant was measured at 649 and 665 nm using a Vernier spectrophotometer, with absolute ethyl alcohol as the blank [24]. To find total chlorophyll content, the equations below were used (Fig. 5); findings of average total chlorophyll content within groups were shown in a column chart.

$$\begin{array}{l} \text{Chlorophyll a} = 13.95A_{665} - 6.88A_{649} \\ \text{Chlorophyll b} = 24.96A_{649} - 7.32A_{665} \\ \text{Total chlorophyll} = \frac{V(\text{Chl a} + \text{Chl b})}{1000 W} \end{array}$$

A = absorbance at a specific wavelength
V = final volume of chlorophyll extract in absolute ethyl alcohol (10 mL)
W = fresh weight of tissue extracted or used for chlorophyll estimation (0.3g)

Figure 5. Equations to find total chlorophyll content; adapted from [24]

2.6.5 Plant height

Plant growth was monitored by measuring the height of the plant from the soil to its tip in centimeters [20]. This was done in order to make sure the growth of the plants was not compromised by PMFC conditions [20]. Plants heights were collected every other day of the experimental period (on days 1, 3, and 5), and depicted in a line graph.

2.6.6 Statistical analyses

To determine if means were significantly different from each other ($p < 0.05$), data was analyzed with t-tests on Google Sheets. Charts and graphs were created in Microsoft Excel for Windows 7.

III. RESULTS

3.1 Voltages

For voltage, a mean of values collected over a period of 1 hour using a voltage probe was obtained every day for 5 days using a Vernier voltage probe and LabQuest. Fig. 6 shows the variation of voltage generation over the experimental period for each group. Generally, the voltages varied without severe fluctuations. However, the Bacillus only group maintained less inconstancy over time. According to the averages of each group, depicted in Fig. 7, the Bacillus only group generated significantly higher voltages than the others, where $p < 0.05$; in other groups, no significant difference was observed. With an average of 36.85 mV, its voltages were 29% higher than the control group, 15% higher than the AMF group, and 15.5% higher than the AMF + Bacillus group.

Variation of voltage generation with time

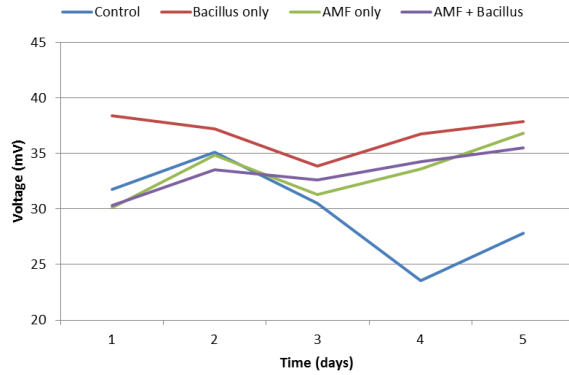


Figure 6. Voltage generation of PMFCs over the course of experimentation; each day, an average of 20 separate values (n=20) was obtained over 1 hour of observation for each group (with one PMFC each).

Average voltages

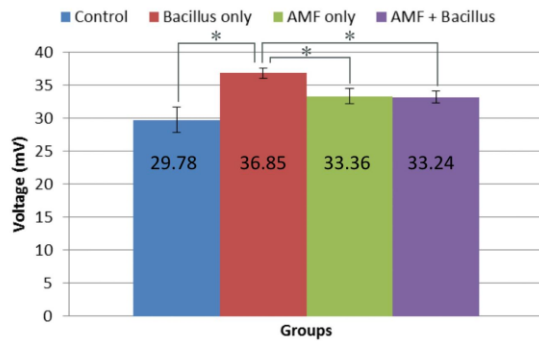


Figure 7. Average voltages, found by taking the mean of the averages observed on each of the 5 days of experimentation (n=5). Error bars represent ± 1 SEM. * represents $p < 0.05$.

3.2 Current/power densities

Current and power densities were obtained by dividing current (voltage divided by resistance) and power (current times voltage) by anode surface area. A similar trend to voltage was observed; the Bacillus only group did significantly better than the others ($p < 0.05$). On average, it produced current densities 24% higher than the control group and 11% higher than the AMF and AMF + Bacillus groups. Its power densities were 51% higher than the control group, 22% higher than the AMF group, and 23% higher than the AMF + Bacillus group. Maximum voltages, power densities, and current densities followed the same pattern. For all groups, power densities were between 400-700 mW/m², while current densities ranged from 15-19 mA/m².

Average power densities

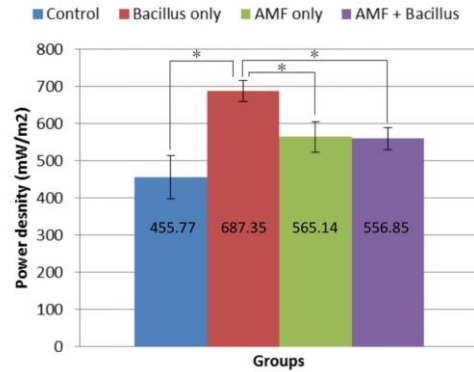


Figure 8. Average power densities, found by taking the mean of the averages observed on each of the 5 days of experimentation (n=5). Error bars represent ± 1 SEM. * represents $p < 0.05$.

Average current densities

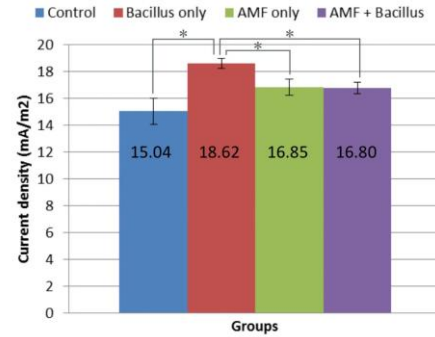


Figure 9. Average current densities, found by taking the mean of the averages observed on each of the 5 days of experimentation (n=5). Error bars represent ± 1 SEM. * represents $p < 0.05$.

	Maximum voltage (mV)	Maximum power density (mW/m ²)	Maximum current density (mA/m ²)
Control group	35.15	624.25	17.76
Bacillus group	38.40	745.03	19.40
AMF group	36.85	686.10	18.62
AMF + Bacillus group	35.49	636.39	17.93

Figure 10. Table of maximum voltages, power densities and current densities, with each value representing the average (n=20) obtained on the day with the highest (respective) value.

3.3 Polarization/power density curves

Polarization and power density curves depict voltage and power density on the y-axis, and current density on the x-axis. They were obtained by collecting voltages with varied external resistances from 100-1000 Ω , with 10 minutes allocated to each resistor. For the polarization curves, there was generally a slight increase in voltage with the increase of current densities at low current densities, followed by a gradual downward slope beginning to show a decrease in voltages due to ohmic losses (internal resistance). Notably, activation losses (a steeper drop in voltage at low current densities) are most prominent in the control and AMF group, which indicate a higher activation energy of the anode reaction. This could be due to the absence of Bacillus, which

conducts the reactions providing the anode with more electrons (rather than those in the soil or released by the plant). For power density curves, power densities increased as current density increased, and did not start decreasing within the domain where data was collected. Interestingly, the AMF group reached higher power and current densities than the others, and experienced less ohmic losses in its polarization curve.

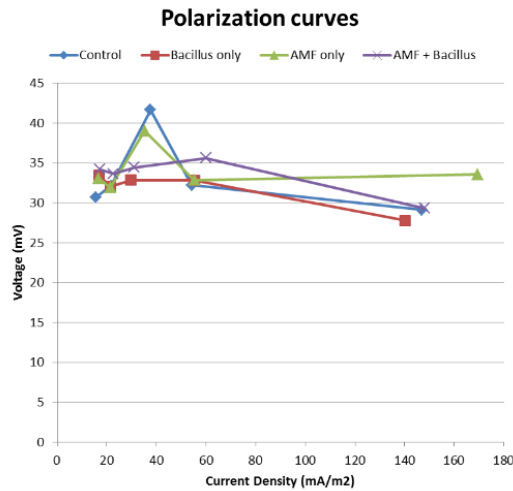


Figure 11. Polarization curves of each group/PMFC. Voltages and current densities collected with each resistor were graphed as points ($n=1$); current density is on the x-axis and voltage is on the y-axis.

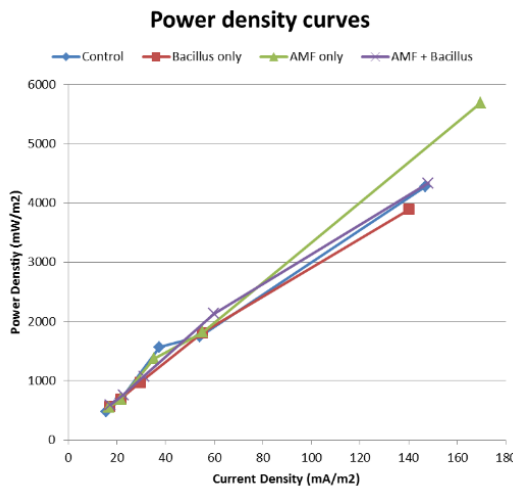


Figure 12. Power density curves of each group/PMFC. Power densities and current densities collected with each resistor were graphed as points ($n=1$); current density is on the x-axis and power density is on the y-axis.

3.4 Plant health

Plant health was kept track of throughout the 5 day experimental period using values of total chlorophyll content (obtained upon the conclusion of the experiment) and plant height (obtained every other day of experimentation). For chlorophyll content, values were only obtained for two groups (both with Bacillus), which did not significantly differ from one another; this was because bacteria was inoculated into the PMFCs without bacteria after preliminary testing, which suggested that increasing the concentration of

bacteria would be beneficial. The new concentrations were put into the PMFCs with no Bacillus because inoculating a more concentrated suspension into soil without an already present concentration (from the last trial) would be more straightforward. Therefore, after experimentation (when it would be appropriate to macerate the leaves), all PMFCs had been inoculated with Bacillus.

As for plant height (which accounted for plant growth), a steady increase was observed in all PMFCs. However, the experimental groups with Bacillus had plants with higher starting heights.

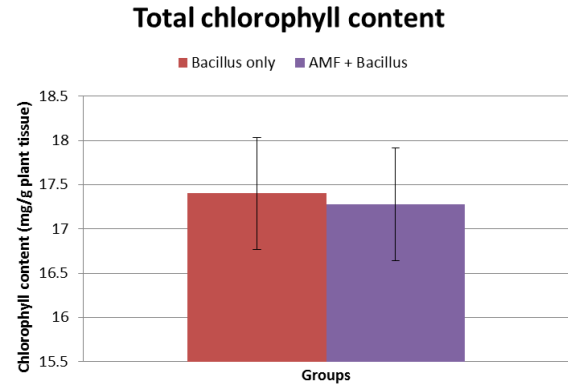


Figure 13. Average chlorophyll contents of groups with Bacillus, found at the end of the experimental period by extracting pigments from macerated leaves and using a spectrophotometer ($n=4$). Error bars represent ± 1 SEM.

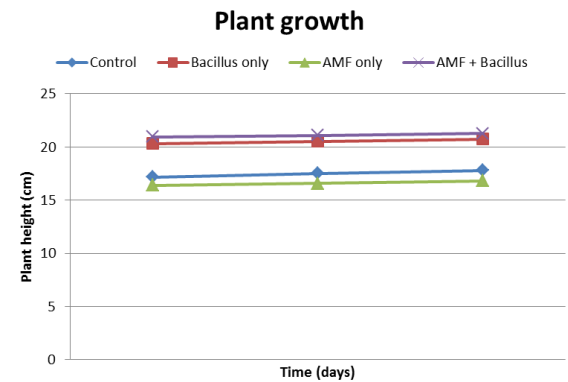


Figure 14. Height of the plant in each PMFC over the course of experimentation, observed on days 1, 3, and 5 of the 5-day experimental period ($n=1$).

IV. DISCUSSION

AMF alone was not associated with a significant increase in electricity generation. Rather, the Bacillus only group produced voltages, current densities, and power densities significantly higher ($p < 0.05$) than those of the control, AMF only, and AMF + Bacillus groups, which were not significantly different from each other.

In particular, the Bacillus group produced power densities greater than that of previous literature and voltages and current densities lower than that of previous literature. Specifically, power densities may have been higher due to the smaller anode surface area (compared to past studies)

utilized in this study; since power density is calculated by dividing the voltage squared by the anode surface area, smaller areas would correspond to a larger power density at a given voltage. While the surface area of this study's graphite rod was 19.79 cm², a study utilizing the same plant and membrane with a different anode [20] measured an anode surface area of about 125 cm². As a result, it had a recorded maximum power density of 5.38 mW/m². Meanwhile, this study's maximum power densities were more than 137 times more than [20]. However, the smaller anode surface area in this study may have negatively impacted voltage, compared to [20] and [23], with voltages of 620 mV and 350 mV, higher than the 36.85 mV produced here. Typically, brush anodes with larger surface areas (implemented in the aforementioned studies) are used to create higher voltages due to having more space for the bacteria to transfer electrons and produce bioelectricity. Similarly, current densities were lower than in previous studies. The maximum current density, found in the *Bacillus* only group, was 19.40 mA/m²; however, in [20], higher current densities were achieved (with a maximum of 38.64 mA/m²), which may have been due to using lower external resistances (as part of the denominator when calculating current density).

The combined group with *Bacillus* and AMF was expected to outperform the others in electricity generation, with the *Bacillus* being present to metabolize root exudates to produce electricity and the AMF being beneficial for root growth (leading to the production of more root exudates). However, the AMF + *Bacillus* group showed no significance compared to the AMF only and the control groups. This may have been due to insufficient nutrient availability for the AMF and *Bacillus subtilis* when together, leading to competition for nutrients. AMF receives 4-20% of carbon from photosynthesis [25], and carbon gets metabolized by electrochemically active bacteria to generate electricity, so AMF likely had a detrimental effect. Together, the AMF and *Bacillus subtilis* may not have been able to successfully carry out their respective roles in the PMFC; thus, without significantly increased electricity generation, AMF may not have been suitable for optimizing PMFC performance with the nutrient level supplied.

In the experiment, there were certain limitations. For example, when first separating *E. aureum* into smaller pieces for each PMFC group, plant root size and biomass were only kept roughly equal across groups. They were not entirely controlled for, and there was a variance in height and root size (plants were labeled as large, medium, or small). This variation may have affected root exudate production because having more biomass is vital for producing more exudates [11] (which are metabolized by EAB to produce bioelectricity), and having a larger root surface area provides more space for EAB to transfer electrons to the anode. However, using different plants across groups did not affect plant health assessed across groups; there was no significance found in chlorophyll content and plant height. Thus, plant health was kept consistent. Throughout the study, plants were not kept in controlled conditions, with a set temperature and light/dark conditions.

They were kept at room temperature, which may have fluctuated and affected growth conditions. Moreover, the amount of light each PMFC plant received was not controlled for through an established light/dark cycle, so this may have affected the growth of the plants in each PMFC (along with exudate production).

Moreover, internal resistance was not factored into the total resistance, unlike previous studies with PMFCs [20]. In other words, not all sources of resistance were accounted for; in addition to the resistor (external), the anode and copper wire also serve as sources of resistance (internal). Also, due to time constraints, ANOVA was not conducted for the chlorophyll content because only one sample could be obtained per group (except those with *Bacillus*). No significant difference was found between the *Bacillus* only and AMF + *Bacillus* groups, but it is unknown whether the AMF only and control groups would have any significance. Additionally, resistors were not soldered to PMFCs when finding the power density and polarization curves. Instead, due to time constraints, tape was used, which did not ensure complete contact between the resistor and the copper wire. As a result of these poor connections, internal resistance may increase, leading to voltage loss.

In future research, additional nutrients should be added to reduce competition between the *Bacillus* and AMF. Due to this competition for nutrients, it is premature to evaluate whether or not AMF as a sole variable is beneficial to PMFCs in general, so more research should be conducted. Moreover, plant biomass and root length should be controlled for. Lastly, future studies should measure internal resistance, as previous studies with PMFCs have done [20], and not just use external resistance (from the resistor) as the sole measure of resistance. In addition to providing more insight into the behavior of the PMFC in general, this would also provide more accurate results when factored into the total resistance.

Future studies should also evaluate the link between chlorophyll content and voltage. Although significance could not be obtained for chlorophyll content in the AMF and control groups, it can be compared to voltage generation with more samples in the future. There may be a link between these variables because chlorophyll allows the plant to absorb sunlight, which helps it carry out photosynthesis. Through this process, plants in PMFCs produce root exudates for the EAB to metabolize and generate electricity. Finally, more efforts should be made to control testing for the polarization and power density curves, as no single resistance proved to benefit electricity production. While 560 ohms seemed beneficial for voltage generation, a range of 300 to 1000 ohms promoted higher power and current densities. In the future, resistors should be soldered (instead of taped) on, ensuring contact between them and the wire and increasing the accuracy of the power densities and voltages. Additionally, a wider range of resistances should be evaluated, because higher current densities (which are obtained with lower external resistances) could not be depicted in the polarization and power density curves.

V. CONCLUSION

This study examined the effect of AMF on PMFCs with/without *Bacillus subtilis*. The results show that the group with *Bacillus* only--with an observed average voltage of 36.85 mV, power density of 687.35 mW/m² (almost three times that of pre-existing PMFC literature), and current density of 18.62 mA/m²--performed significantly better than others, including ones with AMF. This suggests that AMF did not benefit the electricity generation of the PMFC, which could be due to its use of root exudates because they decrease their availability in the rhizosphere (for the *Bacillus* to use). Additionally, plant heights indicate that the health of plants were maintained and not compromised by PMFC conditions.

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